Phan Thanh Tan

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SUMMARY

Final-year Computer Science student majoring in Data Engineering & Big Data. I build data pipelines and ML models end to end, and I report every result against a simple baseline with confidence intervals. Looking for a Data Engineer, Data Analyst or Machine Learning internship.

EDUCATION

Ho Chi Minh City University of Technology (HCMUT), VNU-HCM Oct 2022 – Jun 2027 (expected)
Bachelor of Computer Science, major in Data Engineering and Big Data

EXPERIENCE

MyTegami.win · *Backend Developer Intern* May 2025 – Jul 2025 · Remote

- Built the backend AI generation workflow for a bilingual cover-letter platform with Genkit and Gemini 2.0 Flash (TypeScript).
- Defined Zod schemas so that model outputs arrived in a fixed structure the product could use directly.
- Helped ship support for 14 languages and 5 writing tones.

PROJECTS

NYC Taxi Demand Forecasting GitHub · Live demo
DuckDB, SQL, XGBoost, Streamlit, GitHub Actions

- Built a DuckDB star schema over 376 million public NYC TLC trip records from four services, reading two columns per file over HTTP, with data-quality accounting for every file (93.2% of trips placed on a zone).
- Forecast next-hour pickups for 263 taxi zones with one global XGBoost model; on three held-out months it cut MAE by about 40% against a same-hour-last-week baseline (WAPE 9.6–10.4% vs 16.1–17.6%, bootstrap 95% CIs). Checked for leakage by recomputing the lag features independently in SQL.
- Shipped a 3-page Streamlit app (zone map, weekly patterns, forecast diagnostics) and a CI job that rebuilds one month from source and compares it with the committed data.

Movie Recommender MLOps GitHub
scikit-learn (TruncatedSVD), MLflow on DagsHub, Cloudflare R2, FastAPI, Docker, GitHub Actions

- Built a monthly retraining pipeline on MovieLens 25M with a promotion gate: a challenger replaces the champion in the MLflow registry only if its NDCG@10 is higher with 95% confidence.
- On an untouched month (July 2019, 1,475 users) the champion reached HitRate@10 27.7% vs 22.4% for a popularity baseline (NDCG@10 +0.0121, 95% CI [+0.0044, +0.0193]).
- Served the champion through FastAPI in Docker (no credentials in the image), about 5 ms median latency.

Flight Price Prediction & Quote Assessment GitHub · Live demo
pandas, XGBoost, conformalized quantile regression, Streamlit

- Refactored a course project into a reproducible pipeline: one feature function shared by training and serving (fixing a bug that always predicted with duration 0), duplicates removed before the split, outlier fences fitted on the training set.
- R^2 0.872 and MAE 678 INR on a held-out test set; a calibrated 80% price range covered 78.8% of test prices.
- Same airline and route: one-stop flights cost a median 3,038 INR more than non-stop (95% CI 964–4,410).

SKILLS

Languages	Python, SQL, TypeScript
Data	DuckDB, pandas, star-schema modelling, Parquet, Cloudflare R2
Machine learning	scikit-learn, XGBoost, MLflow, bootstrap confidence intervals, conformalized quantile regression
Delivery	FastAPI, Docker, Streamlit, GitHub Actions, Git

CERTIFICATES

TOEIC Listening & Reading 590 (2023) · HackerRank Software Engineer certificate